
MLET: OpenET Value and an Auditable Reference-Evapotranspiration Outlook

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Abstract

Evapotranspiration is not one forecast target. Reference evapotranspiration (ET_o) describes atmospheric demand for a defined reference surface. Actual evapotranspiration (ET_a) also depends on crop condition, soil water, and management. MLET therefore maintains two evidence paths. The retrospective path tests whether OpenET improves daily actual-ET prediction at field-withheld flux stations. The prospective path defines an auditable 20-day, weather-driven reference- ET_o artifact for Idaho. The reproduced Phase 2 record contains 7,923 station-days from 85 stations. The preregistered comparison is M3 OpenETRidge against B2 WeatherRidge. M3 has mean absolute error (MAE) $0.856 \text{ mm day}^{-1}$ versus $1.514 \text{ mm day}^{-1}$ for B2. This is a $0.658 \text{ mm day}^{-1}$ improvement and a 43.4% reduction. A 2,000-replicate station-blocked bootstrap gives a 95% interval of $[0.399, 0.911] \text{ mm day}^{-1}$. M2 OpenETRecal has the lowest descriptive MAE, $0.781 \text{ mm day}^{-1}$, but it is not the preregistered arm. The prospective path applies the ASCE short-reference equation to archived GEFSv12 members. It stores p10, p50, and p90 for leads 1 through 20. Published USBR AgriMet ETos supplies an independent target. One real issue, one station, and 20 targets pass the source and identity checks. That diagnostic has MAE $1.542 \text{ mm day}^{-1}$, while a prior-year climatology has MAE $0.505 \text{ mm day}^{-1}$. It has only 1 bootstrap cluster and fails the frozen support gate. We make no ET_o skill, field-scale actual-ET, or irrigation claim.

Keywords: evapotranspiration; actual ET; reference ET_o ; OpenET; GEFS; AgriMet; forecast audit; provenance; uncertainty

1 Introduction

Evapotranspiration connects the land surface, the atmosphere, and water management. It also names several different physical quantities. Reference ET_o measures atmospheric demand over a defined reference surface (Allen et al., 1998; ASCE Task Committee, 2005). Actual ET_a is the water flux from the observed crop and soil system. It responds to water stress, crop development, soil condition, and management. A retrospective actual-ET estimate and a prospective reference- ET_o outlook can therefore be individually valid and jointly misleading when they share one label.

Satellite ET products provide broad retrospective information. OpenET joins several remote-sensing models and has been evaluated against in situ ET observations across the contiguous United States (Volk et al., 2023). Weather ensembles provide a different evidence source. They can drive a standardized reference- ET_o equation for future dates, but they do not observe future crop or soil state. The scientific question is not only whether a model has a low error. The question is also whether the target, issue time, baseline, and claim remain unchanged from data acquisition to publication.

This paper reports one completed comparison and one incomplete outlook evaluation. The completed comparison asks whether an OpenET feature improves retrospective daily actual-ET prediction under field withholding. The outlook protocol asks whether archived GEFSv12 weather can support a calibrated 20-day reference- ET_o artifact against independent AgriMet ETos targets. The protocol fails closed when source time, station identity, or evaluation support is incomplete.

The work makes four contributions. First, it defines a target ontology that prevents actual ET, reference ET_o , and conditional crop ET from becoming interchangeable. Second, it reproduces a checksum-bound Phase 2 comparison and reports its preregistered model arm. Third, it implements an issue-time-valid reference- ET_o artifact with quantiles, source receipts, and station-to-grid identity. Fourth, it exposes an incomplete evaluation as a machine-readable state. This last contribution prevents one feasible case from becoming a validation claim.

2 Evidence architecture

Figure 1 gives the governing separation. The left path has historical OpenET values and historical flux labels. The right path has a forecast issue, future local dates, and later AgriMet observations. Results do not cross this boundary.

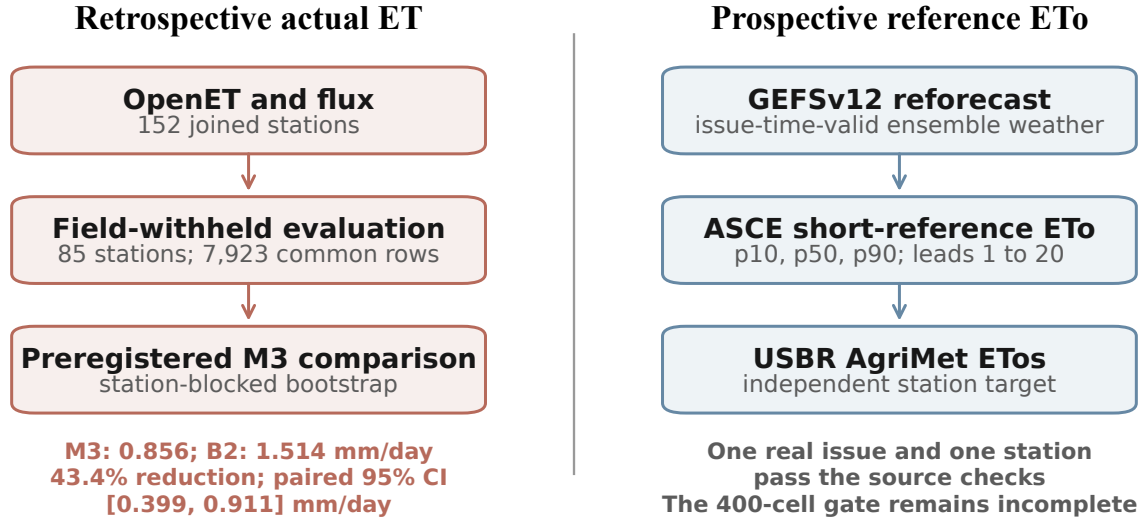


Figure 1. Two targets require two evidence paths. The left result is a retrospective actual-ET comparison. Its fitted-model sample has $n = 7,923$ station-days from 85 stations. The right path is a prospective reference-ET₀ protocol. Its one real case supplies $n = 20$ targets, but the complete 400-cell outcome gate remains open.

2.1 Target ontology

Table 1 states the allowed claims. There is no generic future actual-ET layer in the current product contract. The evaluated future layer is `eto_mm`. Every other future ET quantity needs a separate target and a separate protocol.

Table 1. Target ontology and claim boundary.

Layer or label	Quantity	Time and source	Allowed claim
ET_corr	Energy-balance-corrected actual ET	Historical flux label	Retrospective daily actual-ET prediction
eto_mm	ASCE short-reference ET ₀	Future local date from issue-time GEFS	Regional reference-ET ₀ outlook after the frozen hindcast gate
AgriMet ET ₀ s	Published grass-reference ET	Later station observation; converted from inches per day	Independent reference-ET ₀ evaluation target
potential_et_c_mm	K_c ET ₀ under ample-water assumptions	Conditional future projection	Conditional crop ET only
eta_analysis_mm	Satellite actual-ET analysis	Delayed historical observation	Analysis comparison only
Conditional ET _a scenarios	Named water and soil assumptions	Future scenario	Scenario evidence only

2.2 Proof obligations

Each public claim must bind six identities: the physical target, the source artifact, the forecast issue, the valid date, the independent target, and the evaluation split. A missing identity is not imputed. The evaluator returns an incomplete state. The state can contain diagnostic metrics, but it cannot contain a promotion decision.

3 Data and evidence artifacts

3.1 Retrospective actual-ET data

The Phase 2 build joins daily OpenET model ET with a public flux-tower ET archive on exact station and date keys. The source manifest binds the two archives by identifiers and check-sums. The joined dataset has 16,447 rows from 152 stations. It has 16,444 non-gap labels. The scored label is the energy-balance-corrected ET_corr value. The uncorrected ET column is never scored.

The ridge-model comparison uses the common finite weather subset. This subset has 7,923 station-days from 85 stations. Negative measured ET_corr values remain in the label set. Negative satellite ET and negative reference ET fail input validation. This asymmetric rule keeps observations intact and rejects physically invalid predictors.

The OpenET source data and the flux benchmark follow the Phase II accuracy study (Volk et al., 2023). The present experiment does not re-estimate that study’s site-level accuracy. It asks a narrower prediction question under one repository-frozen split.

3.2 Prospective weather data

The outlook uses the NOAA GEFSv12 weekly reforecast archive (National Centers for Environmental Prediction, 2021). The frozen schedule contains the Wednesday 00Z issues from 2013-01-02 through 2019-12-25. This is 365 planned issues. The feasibility receipt covers one issue, 2019-07-03 00Z. It finds 187 of 187 planned objects and 8,289,206,079 advertised bytes. These are exact inventory values for one issue. They are not a transfer-performance benchmark.

The decoder produces 42,900 weather rows from 195 native grid points, 11 ensemble members, and 20 valid dates. The identity $195 \times 11 \times 20 = 42,900$ is checked before the ET_o builder runs. Each normalized artifact contains its raw-object receipt and SHA-256 digest.

3.3 Independent AgriMet targets

USBR AgriMet publishes daily ET_os as ASCE-EWRI grass-reference ET (U.S. Bureau of Reclamation, 2024a,b). The acquisition adapter parses the official data boundary. It preserves missing markers as exclusions. It converts each non-missing value from inches per day with the exact factor 25.4.

The current station snapshot has 265 stations, including 52 in Idaho. This snapshot does not establish historical coordinates. Historical page evidence supports 19 stations. Their acquisition record contains 31,406 daily values and 115 exclusions from 2015-06-17 through 2019-12-31. The maximum observed station-to-grid distance is 25.475 km across these 19 mappings. The frozen limit is 50 km. This mapping is station identity, not field geometry.

4 Methods

4.1 Phase 2 field withholding

The primary split assigns stations to $k = 10$ deterministic folds. The fold seed is 20260713. A fitted model never sees a row from its held-out stations. Feature means, feature scales, coefficients, and the crop coefficient use training rows only. A separate split trains before 2019-01-01 and describes later rows.

That time split is not the primary decision split.

The experiment fixes six models. B0 persistence uses the previous labeled day at one station. Its 1,555 complete day-pairs make it an oracle-like diagnostic. It is excluded from the H2 decision. B1 learns

$$\widehat{K}_c = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \frac{y_i}{E_i^o}, \quad \widehat{y}_i = \widehat{K}_c E_i^o, \quad (1)$$

where $\mathcal{T}$ contains training rows. B2 is ridge regression on ET_o, mean temperature, vapor-pressure deficit, wind speed, and sine and cosine day-of-year terms. M1 uses OpenET directly. M2 fits an intercept and slope,

$$\widehat{y}_i = \widehat{\alpha} + \widehat{\beta} x_i^{\text{OpenET}}. \quad (2)$$

M3 adds OpenET to all B2 predictors. B2 and M3 solve

$$\widehat{\beta} = \arg \min_{\beta} \left\{ \sum_{i \in \mathcal{T}} (y_i - \mathbf{x}_i^T \beta)^2 + \lambda \|\beta\|_2^2 \right\}, \quad \lambda = 1. \quad (3)$$

The code standardizes predictors with training statistics. It does not penalize the intercept.

4.2 Deterministic metrics and H2

For prediction $\widehat{y}_i$ and observed target y_i , the experiment uses

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\widehat{y}_i - y_i|, \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\widehat{y}_i - y_i)^2}, \quad (5)$$

$$\text{Bias} = \frac{1}{n} \sum_{i=1}^n (\widehat{y}_i - y_i). \quad (6)$$

The preregistration selects the better OpenET-free fitted candidate from B1 and B2. It then compares that model with M3. B2 wins the OpenET-free selection. The paired station-day contrast is

$$\Delta_{\text{MAE}} = \frac{1}{n} \sum_{i=1}^n (|e_{B2,i}| - |e_{M3,i}|), \quad r = \frac{\Delta_{\text{MAE}}}{\text{MAE}_{B2}}. \quad (7)$$

Inference resamples stations, not rows. It uses 2,000 replicates and seed 20260713. The reported interval is the percentile interval of the paired MAE improvement (Efron and Tibshirani, 1993).

4.3 Weather conversion and reference ET_o

The GEFS decoder converts each member to local daily ASCE inputs. Temperature uses $T_C = T_K - 273.15$. Ten-meter wind uses $u_{10} = \sqrt{u^2 + v^2}$ before the reference implementation adjusts its height. Vapor pressure follows

$$e_a = \frac{qp}{\epsilon + (1 - \epsilon)q} \frac{1}{1000}, \quad \epsilon = 0.622, \quad (8)$$

where q is specific humidity in kg kg^{-1} , p is pressure in Pa, and e_a is kPa. Radiation changes from J m^{-2} to MJ m^{-2} . Interval midpoints assign weather intervals to the America/Boise civil day. A UTC-day aggregate fails schema validation.

For each member, the code calls the ASCE short-reference equation (ASCE Task Committee, 2005; Thorp, 2022):

$$\text{ET}_0 = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T+273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)}. \quad (9)$$

Here ET_0 has units mm day^{-1} . The terms have the ASCE daily short-reference definitions. The implementation uses `pyfao56.refet.ascdaily("S", ...)`. The vendored software version is 1.4.3.

Let $E_{g\ell j}$ be Eq. 9 for grid point g , lead ℓ , and member j . The artifact stores

$$q_{g\ell}(\alpha) = Q_\alpha \left(\{E_{g\ell j}\}_{j=1}^m \right), \quad \alpha \in \{0.10, 0.50, 0.90\}. \quad (10)$$

The feasibility case has $m = 11$. The contract rejects duplicate members, missing leads, negative values, and unordered quantiles.

4.4 Issue time and independent target

For issue t_0 and lead ℓ , the valid date is

$$d(t_0, \ell) = \text{date}_{\text{America/Boise}}(t_0) + \ell, \quad \ell \in \{1, \dots, 20\}. \quad (11)$$

Source metadata must show historical availability at or before t_0 . The target must become available only after its completed local day. Retrieval at a later archive date does not change either information boundary.

For station s and date d , the independent target is

$$y_{sd} = 25.4 \text{ ETos}_{sd}^{\text{in/day}}. \quad (12)$$

The code does not recompute this target from GEFS. It does not replace a missing ETos value with zero or interpolation.

4.5 Leakage-safe climatology

The simple baseline is a prior-year station and day-of-year mean. For issue year Y_0 and day of year k ,

$$b_{s,k,Y_0} = \frac{1}{|\mathcal{Y}_{s,k,Y_0}|} \sum_{Y \in \mathcal{Y}_{s,k,Y_0}} y_{s,k,Y}, \quad \mathcal{Y}_{s,k,Y_0} = \{Y : Y < Y_0\}. \quad (13)$$

The evaluated year and all future years are absent. A missing prior-year set creates an explicit exclusion.

4.6 Probabilistic metrics and support

The outcome protocol reports p50 MAE, RMSE, and bias. It also reports p10-p90 coverage and width:

$$C_{10,90} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{q_{0.10,i} \leq y_i \leq q_{0.90,i}\}, \quad (14)$$

$$W_{10,90} = \frac{1}{n} \sum_{i=1}^n (q_{0.90,i} - q_{0.10,i}). \quad (15)$$

Table 2. Field-withheld Phase 2 metrics in mm day^{-1} .

Model	MAE	RMSE	Bias	n
B0 Persistence	0.350	0.572	0.058	1,555
B1 CropCoefficient	1.532	2.005	0.149	7,923
B2 WeatherRidge	1.514	2.687	-0.098	7,923
M1 OpenETDirect	0.784	1.066	0.154	7,923
M2 OpenETRecal	0.781	1.060	0.005	7,923
M3 OpenETRidge	0.856	1.386	-0.013	7,923

For quantile level α , pinball loss is

$$\rho_\alpha(y, q) = \begin{cases} \alpha(y - q), & y \geq q, \\ (\alpha - 1)(y - q), & y < q. \end{cases} \quad (16)$$

The mean score averages Eq. 16 over observations and the three stored quantiles (Koenker and Bassett, 1978; Gneiting and Raftery, 2007). It is not the continuous ranked probability score.

Spatial folds use one-degree blocks. The fold is the SHA-256 digest of a fixed namespace and block, modulo 5. The evaluator crosses 20 leads, four seasons, and five folds. This gives 400 cells. Each cell needs at least 30 paired targets. Thus the minimum support count is 12,000 cell observations. The paired ETo interval uses 1,000 cluster-bootstrap replicates with seed 20260731. A cluster is one local issue date and target station. An interval requires at least two clusters.

5 Results

5.1 Preregistered OpenET value result

Figure 2 and Table 2 report the committed field-withheld result. The common fitted-model sample is $n = 7,923$ station-days from 85 stations. The preregistered H2 model is M3, not M2.

M3 has MAE $0.856 \text{ mm day}^{-1}$. B2 has MAE $1.514 \text{ mm day}^{-1}$. The paired improvement is $0.658 \text{ mm day}^{-1}$, or 43.4%. The 2,000-replicate station-blocked 95% interval is $[0.399, 0.911] \text{ mm day}^{-1}$. The interval excludes zero under the frozen protocol.

M2 has the lowest point MAE, $0.781 \text{ mm day}^{-1}$. Its direct point reduction from B2 is 48.4%, computed on $n = 7,923$ common rows. This is a descriptive contrast. The result record contains no paired M2 interval. We do not attach the M3 interval to M2.

The Phase 2 result is retrospective. It does not test GEFS, the 20-day horizon, soil-water dynamics, or an irrigation decision.

5.2 Real reference-ETo feasibility diagnostic

The one-issue candidate passes the source count, member count, lead count, quantile order, local-day, station-history, target-time, and grid-distance checks. Figure 3 shows its BOII trajectory. The case has one 2019-07-03 issue, one station, and $n = 20$ daily targets.

The p50 MAE is $1.542 \text{ mm day}^{-1}$. The climatology MAE is $0.505 \text{ mm day}^{-1}$. The paired point improvement is $-1.037 \text{ mm day}^{-1}$, so the candidate is worse for this case. The p50 RMSE is $1.691 \text{ mm day}^{-1}$, and the bias is $1.542 \text{ mm day}^{-1}$.

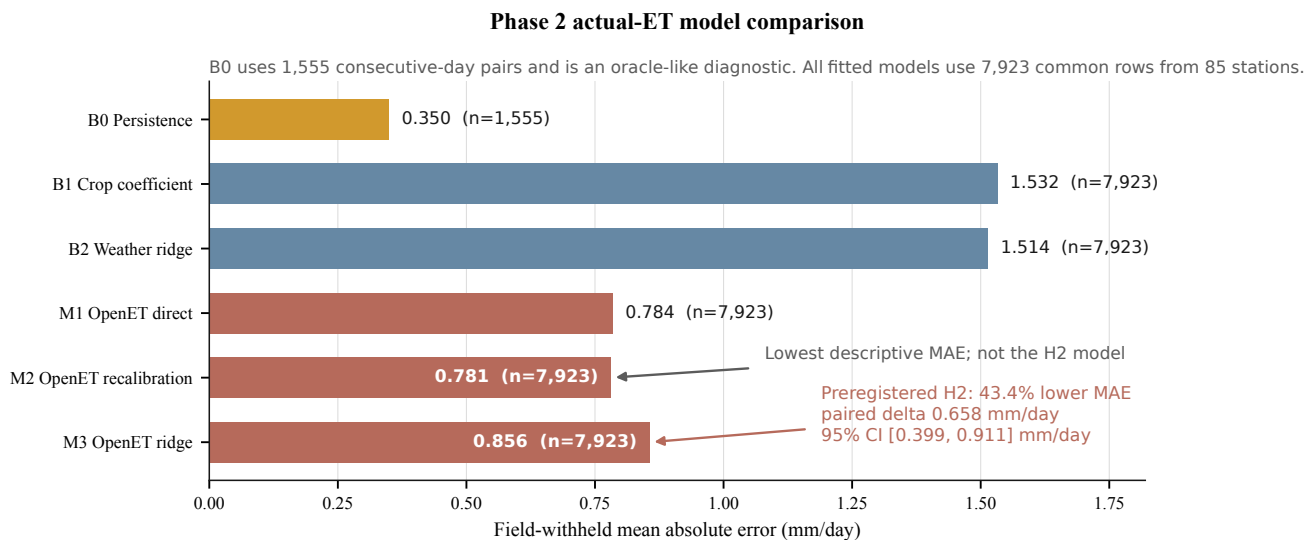


Figure 2. Field-withheld actual-ET MAE. B2 and M3 form the preregistered H2 comparison. Their common sample contains $n = 7,923$ rows from 85 stations. M2 has the lowest descriptive point MAE, but it does not inherit the M3 interval. B0 uses $n = 1,555$ consecutive-day pairs and is excluded from H2.

The p10-p90 coverage is 0.20, against a nominal width of 0.80. The mean interval width is $1.805 \text{ mm day}^{-1}$. The mean three-quantile pinball loss is $0.540 \text{ mm day}^{-1}$.

These values have no sampling interval. The case has only 1 bootstrap cluster. It also has one target per occupied lead cell, below the frozen minimum of 30. The result is a negative feasibility diagnostic, not an estimate of general forecast skill.

5.3 Spatial artifact and support state

Figure 4 shows the real native-grid artifact. Each panel contains 195 weather-grid points. It does not show field boundaries. The gold marker identifies BOII and its mapped grid point.

Figure 5 makes the incomplete state explicit. The real archive supplies one target in each JJA, fold-4 lead cell. This is 20 cell observations. The frozen gate needs 12,000 cell observations when every one of the 400 cells reaches its minimum. The present archive therefore supports no full-hindcast claim.

6 Discussion

6.1 What the Phase 2 result establishes

The completed result supports one narrow conclusion. Under the frozen field-withheld split, M3 OpenETRidge reduces retrospective daily actual-ET MAE against B2 WeatherRidge. The paired interval accounts for station clustering. It does not account for a new geographic domain, a new label system, or a future weather issue. The comparison therefore supports OpenET value within this benchmark. It does not validate the outlook.

M2 creates an important reporting test. Its point MAE is lower than M3. A reader could reasonably call M2 the descriptive winner. The preregistered H2 arm is still M3 because the runner fixed that model. A paper must report both facts. Combining M2's point error with M3's interval would create a result that no execution produced.

The persistence row has a much lower MAE than every fitted model. That value uses only 1,555 consecutive-day pairs and consumes the previous observed label. It is not available for a lead-1 forecast without an observed prior day. We therefore treat it as an oracle-like diagnostic, not the H2 baseline.

6.2 What the feasibility case establishes

The reference-ET₀ case shows that the data path can produce an auditable artifact. The path starts with an archived issue and ends with a later, independent station target. It preserves local-day semantics, members, quantiles, source receipts, and checksums. This is a systems result.

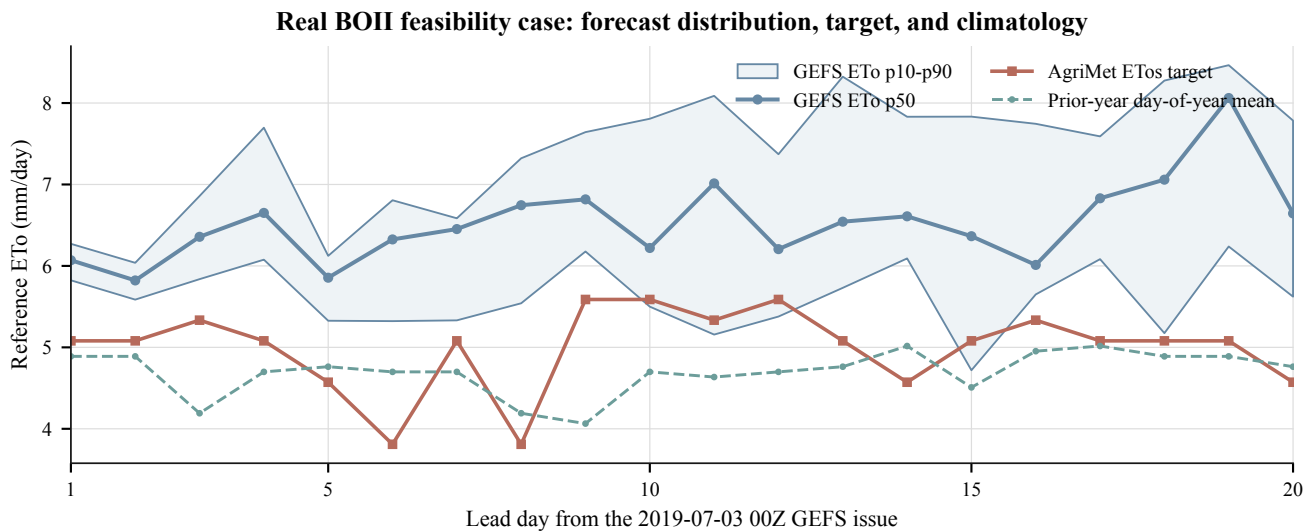
The diagnostic forecast is worse than the simple climatology for the one case. Its median forecast also has a positive bias of $1.542 \text{ mm day}^{-1}$. The p10-p90 interval covers only 20% of the 20 targets. These numbers warn against promotion. They do not identify a stable lead effect, season effect, or statewide effect. One issue and one station cannot estimate those effects.

The incomplete state is scientifically useful. It prevents source feasibility from becoming forecast validation. It also preserves the negative diagnostic instead of deleting it. A full result must include every frozen support cell, even when the candidate does not improve on climatology.

6.3 Why target separation matters

Reference ET₀ is a weather-derived demand index. Actual ET is a realized surface flux. An actual-ET outlook also needs crop state, root-zone water, and management assumptions. A crop coefficient can convert ET₀ to potential crop ET under named assumptions. It cannot by itself create unconditional actual ET.

This boundary affects model assessment. A retrospective satellite feature can help predict historical actual ET. That evidence does not make the feature available at a future issue. An AgriMet ETos target can assess future reference ET₀. That ev-



Diagnostic only: 1 issue, 1 station, $n=20$ targets. Forecast MAE 1.542; climatology MAE 0.505 mm/day. p10-p90 coverage 0.20; mean width 1.805 mm/day. Support $20 < 30$ and one bootstrap cluster, so no skill interval is identified.

Figure 3. One real BOII feasibility case. The GEFS member distribution gives p10, p50, and p90 reference ET_0 for leads 1 through 20. The independent target is AgriMet ETos. The baseline is the prior-year station and day-of-year mean. The scope is one issue, one station, and $n = 20$ targets. It is diagnostic only.

idence does not assess field-scale actual ET. The ontology in Table 1 keeps these claims separate by construction.

7 Limitations

The paper has seven main limitations.

1. The complete 365-issue GEFS and AgriMet outcome archive is not present. Reference- ET_0 skill remains **evaluation pending**.
2. The current AgriMet registry is not a historical movement ledger. Historical page evidence covers 19 stations.
3. The BOII location evidence brackets the case date. It does not prove an unchanged coordinate for every date from 2013 through 2019.
4. The spatial artifact contains weather-grid points. It does not contain field boundaries, crop labels, or area weights.
5. The Phase 2 comparison uses public flux labels and historical OpenET. It does not measure future forecast performance.
6. The one-case ET_0 metrics have $n = 20$ and one cluster. Their uncertainty is not estimable under the frozen bootstrap.
7. The codebase contains a non-serving FAO-56 dual water-balance scaffold. No result in this paper depends on that untrained scaffold.

8 Conclusions

MLET separates retrospective actual ET from prospective reference ET_0 . The Phase 2 record supports a preregistered M3 versus B2 actual-ET result. M3 reduces MAE by 43.4%, with a paired 95% interval of $[0.399, 0.911]$ mm day⁻¹. M2 has a lower descriptive point MAE, but it has no recorded paired interval.

The reference- ET_0 path produces a real, checksum-bound

candidate and an independent target. Its one-case diagnostic is worse than climatology and has insufficient support. The paper therefore makes no reference- ET_0 skill or irrigation claim. Promotion requires the complete frozen hindcast and a separate release review.

Data and code availability

The repository stores the Phase 2 result, its independent reproduction receipt, the GEFS feasibility receipts, the AgriMet target artifacts, and the schema-v4 feasibility archive. The manuscript figures read those artifacts directly. The scripts `scripts/build_arxiv_claims.py` and `scripts/build_arxiv_figures.py` regenerate the numerical macros and figures. The complete raw GEFS plan is approximately 3.026 TB by a single-issue multiplication estimate. That estimate uses 365 planned issues and one 8,289,206,079-byte inventory. It has no sampling uncertainty and is not a measured full transfer.

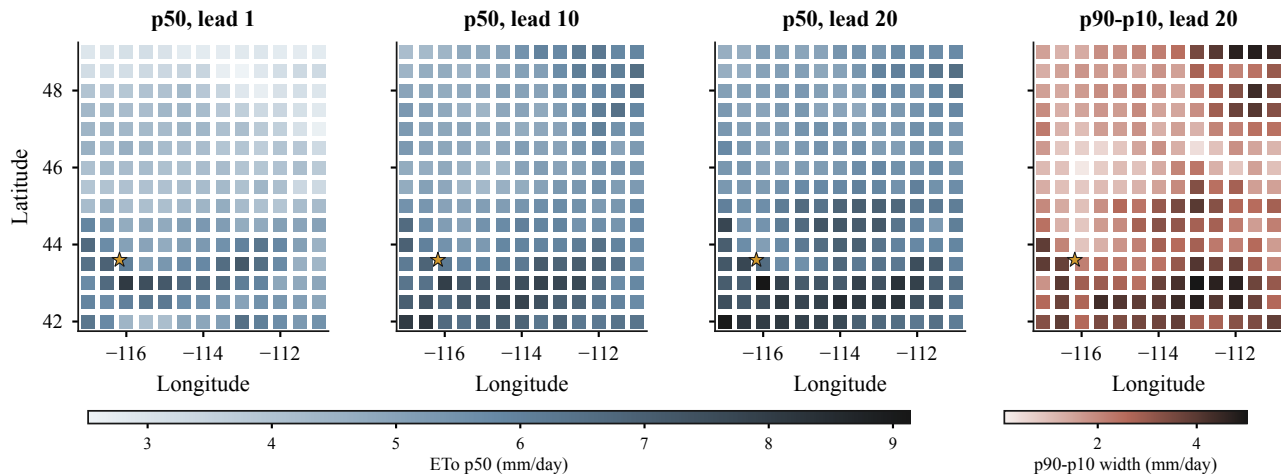
Reproducibility statement

The Phase 2 protocol uses seed 20260713 and 2,000 station-blocked replicates. The ET_0 protocol uses seed 20260731 and 1,000 issue-station cluster replicates. Generated claims fail when B2 minus M3 does not equal the H2 delta. They also fail when M2 is not the descriptive minimum or when the ET_0 evidence scope changes. Appendix B records the exact artifact digests used here.

A Implementation map

Table 3 maps each manuscript operation to the local implementation. This map is part of the claim audit. It is not a substitute for the source code.

Native GEFS grid in the real 2019-07-03 research candidate

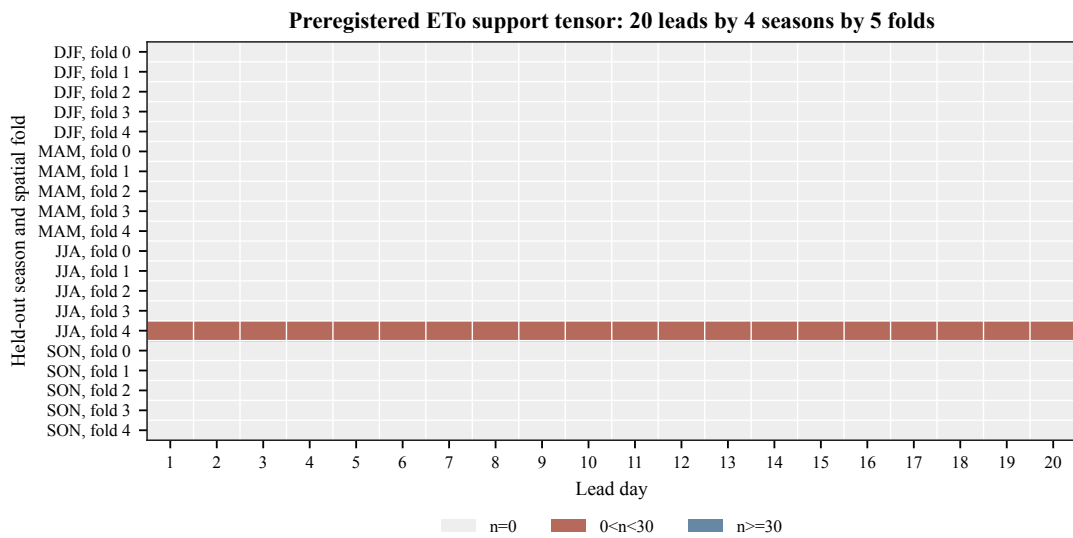


Each panel contains 195 weather-grid points. The gold marker is BOII. The points are not field boundaries or area-weighted state estimates.

Figure 4. Native GEFS reference-ET₀ fields for the real 2019-07-03 candidate. The first three panels show p50 at leads 1, 10, and 20. The last panel shows p90 minus p10 at lead 20. Each panel contains $n = 195$ grid points. No field geometry or area-weighted Idaho estimate is inferred.

Table 3. Equation and claim map to the implementation.

Operation	Local implementation	Enforced invariant
Phase 2 join	<code>src/mlet/build_dataset.py</code>	Exact station and date key; ET_corr label
Model definitions	<code>src/mlet/baselines.py</code>	Training-only fit; ridge penalty 1.0
Blocked H2 interval	<code>src/mlet/evaluate.py</code>	Station blocks; 2,000 replicates
Phase 2 runner	<code>src/mlet/experiments/phase2_openet_value.py</code>	H2 arm is M3 versus best B1 or B2
GEFS decode	<code>src/mlet/sources/gefs_reforecast.py</code>	Local day; member and component identity
Reference ET ₀	<code>src/mlet/outlook/eto.py</code>	ASCE short reference; p10, p50, p90
Candidate contract	<code>src/mlet/outlook/eto_build.py</code>	20 leads; research status; no extra layers
AgriMet target	<code>src/mlet/sources/agrimet.py</code>	Independent ETos; exact 25.4 conversion
Baseline builder	<code>scripts/build_eto_target_index.py</code>	Prior years only for station and day of year
Outcome evaluator	<code>src/mlet/outlook/eto_hindcast.py</code>	400 cells; fail-closed completion state



The real feasibility archive supplies one target in each JJA-fold-4 lead cell: 20 cell observations. The frozen minimum is 12,000 cell observations (400 cells x 30).

Figure 5. Frozen reference-ET_o support tensor. The real feasibility archive occupies 20 cells with one target each. No cell reaches the $n \geq 30$ threshold. The complete protocol requires all 400 lead-season-fold cells.

B Artifact ledger

The following digests bind the plotted evidence. Paths are relative to the repository root.

Phase 2 result

docs/results/phase2_openet_value.json

SHA-256: 2a2cece4a4febaa6105d5182ae01e161b98db7c18bb0ae499310fe160cd5eee3

ETo feasibility evidence

data/outlook/eto_feasibility_archive/evidence.json

SHA-256: 890555ea6e0f912655fbdcb7cd1291c019ce9545e21407f3a9432f1582bba38

ETo evaluation

Deterministic evaluation SHA-256: 9735fedb7a11c6aa18c5fa4146288acda5d487ba89bb0df8ad6efc414bccca039

BOII outlook

data/outlook/eto_feasibility_archive/cases/issue-20190703-station-BOII-season-JJA-fold-4/outlook.json

SHA-256: 9476bab0be5cd879d1912a40c73ff5d694d9ae95502d67e8f6d82ee2426adb3b

BOII target

data/outlook/eto_feasibility_archive/cases/issue-20190703-station-BOII-season-JJA-fold-4/target.json

SHA-256: 7aabbf02699745d7051bc09d7c00bd59e969abbcf639d391728086fd817d9751

C Claim ledger

Table 4. Publication claims and their current states.

State	Claim	Evidence boundary
Supported	M3 reduces retrospective actual-ET MAE against B2 by 43.4%.	$n = 7,923$, 85 stations, 2,000 station-blocked replicates, 95% interval [0.399, 0.911] mm day ⁻¹ .
Supported	M2 has the lowest descriptive MAE at 0.781 mm day ⁻¹ .	$n = 7,923$ common rows; no paired M2 interval in the record.
Diagnostic	One BOII ETo case has MAE 1.542 mm day ⁻¹ .	One issue, one station, $n = 20$, one bootstrap cluster.
Pending	The 20-day reference-ET _o outlook improves on climatology.	Full 365-issue archive, all 400 cells, and release review required.
Out of scope	The artifact predicts field-scale actual ET or irrigation need.	No field target, crop state, soil-water target, or irrigation outcome exists.

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